

Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued. . .]

- Appendix on s -wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Theoretical introduction

[To be continued...]

1.1 Antiferromagnetic ordering in the Hubbard model

Consider the ordinary Hubbard model:

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad t, U > 0 \quad (1.1)$$

The two competing mechanisms are site-hopping of amplitude t and local repulsion of amplitude U . For this model defined on a bipartite lattice at half filling and fixed electron number, it is well known [hirsch1985hubbard] that, below a certain critical temperature T_c and above some (small) critical repulsion U_c/t , the ground-state acquires antiferromagnetic (AF) long-range ordering, schematically depicted in Fig. 1.1a. The mechanism for the formation of the AF phase takes advantage of virtual hopping, as described in App. ??; the Mean-Field Theory (MFT) treatment of ferromagnetic-antiferromagnetic orderings in 2D Hubbard lattices is discussed in App. ??.

In this chapter the discussion is limited to the two-dimensional square lattice Hubbard model. The lattice considered has L_{ℓ} sites on side $\ell = x, y$, thus a total of $L_x L_y$ sites. The total number of single electron states is given by $D = 2L_x L_y$. All theoretical discussion neglects border effects, thus considering $D \rightarrow +\infty$.

1.2 The Extended Fermi-Hubbard model

The Extended Fermi-Hubbard model is defined by:

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \quad (1.2)$$

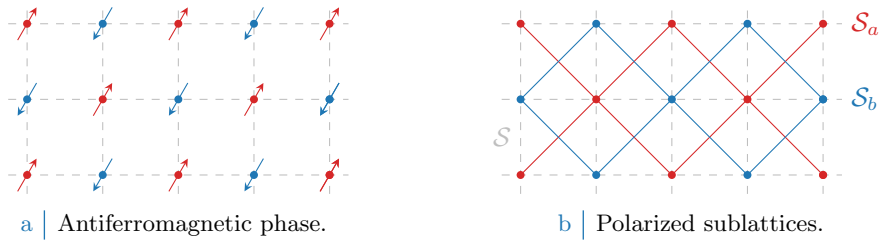


Figure 1.1 | Schematic representation of the AF phase. Fig. 1.1a shows a portion of the square lattice with explicit representation of the spin for each site. Fig. 1.1b divides the square lattice $\mathcal{S}$ in two polarized sublattices $\mathcal{S}_a, \mathcal{S}_b$. The AF phase results from the interaction of two inversely polarized “ferromagnetic” square lattices.

Note that, on a square lattice, we can perform the summation over NN just as

$$\sum_{\langle ij \rangle} \equiv \sum_{i \in \mathcal{S}_a} [\delta_{j=i+\delta_x} + \delta_{j=i-\delta_x} + \delta_{j=i+\delta_y} + \delta_{j=i-\delta_y}]$$

where notation of Fig. 1.1b has been used. The last term represents an effective attraction between neighboring electrons, of amplitude V . Such an interaction is believed [cao2025p-wave] necessary to describe the insurgence of high- T_c superconductivity in cuprate SCs. [To be continued...]

1.2.1 Experimental insight on NN attraction

[Add:

- High T_c SC in cuprates;
- Experimental evidence of topological SC;
- Insertion of the non-local attraction;

]

Chapter 2

Introduction to Hartree-Fock methods

The Hartree-Fock (HF) method is a variational approach to determine the ground-state wavefunction of a many-body hamiltonian. The wavefunction is searched in a variational scheme as the one minimising energy, limiting search to a specific subspace of states. The general idea is to approximate the real hamiltonian with another non-interacting fermionic hamiltonian. The new operator should be quadratic in terms of fermionic operators.

First, we need to define what Hilbert subspace we work with. Then we need to identify a way of parametrising the possible non-interacting hamiltonians and their respective ground-states, and finally a method to choose the best approximation to our original model. This is done in next sections. The present discussion is largely based on [6, 8].

2.1 Prerequisites and generalities

Let us start by discussing the properties of the many-body states we wish to use to approximate the real ground-state of the system. We introduce Slater determinants to grasp the general idea behind HF variational search, and then extend to a larger set of states.

2.1.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is “to antisymmetrize” a set of orthonormal N one-body spin-orbitals, being N the total number of electrons and D the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma} \int d\mathbf{r} \psi_i(\mathbf{r}, \sigma) \psi_j(\mathbf{r}, \sigma) = \delta_{ij}$$

being $\mathcal{V}$ the total volume. Let then the antisymmetrization operator

$$\mathcal{A} \equiv \frac{1}{\sqrt{N!}} \sum_{\mathbf{P} \in \mathcal{S}_N} (-1)^{\sigma(\mathbf{P})} \mathbf{P}$$

being $\mathcal{S}_N$ the $N!$ elements symmetric group of N objects permutations, and $\sigma(\mathbf{P})$ the “parity factor” of any permutation $\mathbf{P} \in \mathcal{S}_N$. $\sigma(\mathbf{P})$ counts the number of exchanges needed to achieve a given configuration. This way, an even number of exchanges gives $+1$, an odd number gives -1 . The permutation operator exchanges spin-coordinates couples,

$$\mathbf{P} : (\mathbf{r}, \sigma) \leftrightarrow (\mathbf{r}', \sigma')$$

thus any given pair is exchanged with another.

Equipped with $\mathcal{A}$, we define the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \dots; \mathbf{r}_N, \sigma_N) = \mathcal{A} \left\{ \prod_{i=1}^N \psi_i(\mathbf{r}_i, \sigma_i) \right\}$$

which by construction is antisymmetric under exchange of coordinates and spins pairs. This class of wavefunctions, obtained as antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \dots; \mathbf{r}_N, \sigma_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(\mathbf{r}_1, \sigma_1) & \psi_1(\mathbf{r}_2, \sigma_2) & \cdots & \psi_1(\mathbf{r}_N, \sigma_N) \\ \psi_2(\mathbf{r}_1, \sigma_1) & \psi_2(\mathbf{r}_2, \sigma_2) & \cdots & \psi_2(\mathbf{r}_N, \sigma_N) \\ \vdots & \vdots & & \vdots \\ \psi_N(\mathbf{r}_1, \sigma_1) & \psi_N(\mathbf{r}_2, \sigma_2) & \cdots & \psi_N(\mathbf{r}_N, \sigma_N) \end{bmatrix}$$

Determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital ψ_i can be doubly occupied (which is, all spin-coordinate couples must be different). Pauli exclusion principle is included in this framework.

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1, \psi_2, \dots, \psi_N\}$ to approximate the true ground-state of the model. The space of possible ‘‘Slater-like’’ many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to a couple of annihilation-creation fermionic operators,

$$\psi_i \rightarrow \hat{c}_i, \hat{c}_i^\dagger$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^N \hat{c}_i^\dagger |\Omega\rangle \quad (2.1)$$

being $|\Omega\rangle$ the vacuum.

2.1.2 Wick’s theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick’s theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (2.1) for a given set of electronic operators $\hat{c}_i, \hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < H$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (2.1) (thus, non populated). Then it holds:

$$\begin{aligned} 0 \leq i \leq N & : \hat{c}_i^\dagger |\Psi\rangle = 0 \\ i > N & : \hat{c}_i |\Psi\rangle = 0 \end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases} \hat{c}_i^\dagger & \text{for } 0 \leq i \leq N \\ \hat{c}_i & \text{for } i > N \end{cases}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$. They are annihilation and creation operators above the ‘‘vacuum’’ state $|\Psi\rangle$. In other words, any Slater state like Eq. (2.1) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^D \left(u_i^{(\ell)} \hat{c}_i + v_i^{(\ell)} \hat{c}_i^\dagger \right) \quad (2.2)$$

for some coefficients $\{u_i^{(\ell)}, v_i^{(\ell)}\}$. Consider the product operator:

$$\hat{\mathcal{O}} \equiv \hat{O}^{(1)} \hat{O}^{(2)} \dots \hat{O}^{(2k)}$$

for $k \in \mathbb{N}$. Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{\mathcal{O}} = \dots + (\text{coefficient}) \times \hat{C}_1 \hat{C}_2 \dots \hat{C}_{2k} + \dots$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some j . Now, Wick's theorem states that

$$\begin{aligned} \hat{C}_1 \hat{C}_2 \dots \hat{C}_{2k} &= N \left\{ \hat{C}_1 \hat{C}_2 \dots \hat{C}_{2k} \right\} \\ &+ \sum_{(\alpha\beta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} N \left\{ \prod_{i \neq \alpha, \beta} \hat{C}_i \right\} \\ &+ \sum_{(\alpha\beta)(\gamma\delta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} (-1)^{\zeta_{\gamma\delta}} D_{\gamma\delta} N \left\{ \prod_{i \neq \alpha, \beta, \gamma, \delta} \hat{C}_i \right\} \\ &+ \dots \end{aligned}$$

where $N\{\dots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha \hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are defined as expectation values over $|\Psi\rangle$. Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: if we change basis and express all operators in terms of $\hat{a}$, $\hat{a}^\dagger$,

$$\hat{C}_i \rightarrow \hat{A}_i$$

the above decomposition holds identically, with slight change of notation. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle \Psi | N\{\hat{A}_i \hat{A}_j \dots\} | \Psi \rangle = 0$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\langle \hat{C}_1 \hat{C}_2 \dots \hat{C}_{2k} \rangle = \sum (\text{total sign factor}) \times (\text{product of complete contractions})$$

where sum is intended, implicitly, over all possible ways of choosing k couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. **This is the defining property of gaussian states.** In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

2.1.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. **However, in order to treat some physical phenomena we need to**

extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of “standard” electronic operators.

In this section we extend the notions introduced before and set grounds for the generalised Hartree-Fock-Bogoliubov (HFB) method we use in this thesis.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{c} \\ \hat{c}^\dagger \end{bmatrix}$$

being $\hat{c}$ the vector of all annihilation operators, ordered, and $\hat{c}^\dagger$ of all creation operators. Let us assume the vectors have D entries. Consider an unitary map M , such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\gamma} \\ \hat{\gamma}^\dagger \end{bmatrix}$$

The new Nambu vectors $\hat{\gamma}$ and $\hat{\gamma}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix}$$

If M is unitary, it follows:

$$\begin{aligned} \mathbb{I}_{2D \times 2D} &= MM^\dagger \\ &= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U & U^\dagger \\ V^\dagger & V \end{bmatrix} \\ &= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ (U^\dagger)^2 + (V^\dagger)^2 & U^\dagger U + V^\dagger V \end{bmatrix} \end{aligned}$$

thus, among other conditions, $UU^\dagger + VV^\dagger = \mathbb{I}_{D \times D}$.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^D U_{ij} \hat{c}_j + \sum_{j=1}^D V_{ij} \hat{c}_j^\dagger \quad \hat{\gamma}_i^\dagger \equiv \sum_{j=1}^D U_{ji}^* \hat{c}_j^\dagger + \sum_{j=1}^D V_{ji}^* \hat{c}_j$$

and substituting in the anti-commutator

$$\begin{aligned} \{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^D U_{ii'} \sum_{j'=1}^D U_{j'j}^* \{\hat{c}_{i'}, \hat{c}_{j'}^\dagger\} + \sum_{i'=1}^D V_{ii'} \sum_{j'=1}^D V_{j'j}^* \{\hat{c}_{i'}, \hat{c}_{j'}\} \\ &= \sum_{i'=1}^D U_{ii'} \sum_{j'=1}^D U_{j'j}^* \delta_{i'j'} + \sum_{i'=1}^D V_{ii'} \sum_{j'=1}^D V_{j'j}^* \delta_{i'j'} \\ &= [UU^\dagger + VV^\dagger]_{ij} \\ &= \delta_{ij} \end{aligned}$$

being M unitary.

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i |\Psi\rangle = 0$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \rangle$$

We can apply the result obtained above. Indeed, inverse maps linking old and new operators read:

$$\hat{c}_i \equiv \sum_{j=1}^D U_{ij} \hat{\gamma}_j + \sum_{j=1}^D U_{ji}^* \hat{\gamma}_j^\dagger \quad \hat{c}_i^\dagger \equiv \sum_{j=1}^D V_{ji}^* \hat{\gamma}_j^\dagger + \sum_{j=1}^D V_{ij} \hat{\gamma}_j$$

In Sec. 2.1.2 we argued that, when averaging a product of operators like the one of Eq. (2.2), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \rightarrow \hat{C}_\ell$ and $\hat{c}_i, \hat{c}_i^\dagger \rightarrow \hat{\gamma}_i, \hat{\gamma}_i^\dagger$. Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \gamma_i^\dagger$ obey Fermi anticommutation rules and “see” $|Psi\rangle$ as a vacuum state.

As in Sec. 2.1.2, we conclude that:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}}$$

The subscripts indicate the name we will give to these terms throughout the text. Note that, differently from “pure Slater states” EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov (HFB). For simplicity we will stick, from now on, to the name “HF method” referring to this discussion. **This argument has been carried out at zero temperature, but can be extended to finite temperature using appropriate path-integral techniques.**

2.2 Sketch of HF variational method

Consider the generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha, \beta} A_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} B_{\alpha\beta\gamma\delta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta$$

where

$$A_{\alpha\beta} = A_{\beta\alpha}^* \quad B_{\alpha\beta\gamma\delta} = B_{\delta\gamma\beta\alpha}^*$$

to ensure hermiticity. Moreover, the model must show the following symmetry:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \quad (2.3)$$

because exchanging the position of the two annihilations and the two creations, the operator must remain the same.

Our aim is to approximate the above hamiltonian with something like:

$$\hat{H} \simeq \hat{H}_{\text{HF}} \equiv \sum_{\alpha, \beta} k_{\alpha\beta} \hat{\gamma}_\alpha^\dagger \hat{\gamma}_\beta$$

Old and new operators must be linked by an unitary transform: with identical notation as in previous sections, $\hat{\Gamma} = M\hat{\Phi}$. To find the best approximation to the original hamiltonian means to identify both the γ operators and the k coefficients.

2.2.1 Properties of the target quadratic hamiltonian

Let us enumerate briefly some properties of the target hamiltonian. We will use also the following notation:

$$\hat{a}_\alpha^- \equiv \hat{c}_\alpha \quad \hat{a}_\alpha^+ \equiv \hat{c}_\alpha^\dagger$$

Due to linearity of the unitary mapping, $\hat{H}_{\text{HF}}$ can be expressed also as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\text{HF}} = \sum_{\alpha, \beta} \sum_{p, q} h_{\alpha\beta}^{pq} \hat{a}_\alpha^p \hat{a}_\beta^q$$

We moved all our ignorance into the new $h_{\alpha\beta}^{pq}$ fields.

We can infer some mathematical properties of the fields $h_{\alpha\beta}^{pq}$. First, note that the hamiltonian can be represented through Nambu spinors as

$$\hat{H}_{\text{HF}} = \begin{bmatrix} \hat{c} \\ \hat{c}^\dagger \end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+} \end{bmatrix} \begin{bmatrix} \hat{c} \\ \hat{c}^\dagger \end{bmatrix}$$

From hamiltonian hermiticity we conclude that:

$$\hat{H}_{\text{HF}}^\dagger = \hat{H}_{\text{HF}} \implies h^{++} = [h^{--}]^\dagger$$

which reads, element-wise:

$$h_{\alpha\beta}^{++} = (h_{\beta\alpha}^{--})^* \quad (2.4)$$

Same-sign off-diagonal fields are linked.

Moreover, any hamiltonian can be arbitrarily chosen to be traceless: Physics is not affected by constant energy shifts. Thus we can set

$$\text{Tr } h^{+-} + \text{Tr } h^{-+} = 0$$

Our variational argument does not depend neither on the trace of the fields nor the trace of the original hamiltonian. From this we conclude that also on-diagonal terms are linked: indeed, considering the explicit form of the hamiltonian:

$$\begin{aligned} \hat{H}_{\text{HF}} &= \sum_{\alpha,\beta} \left[h_{\alpha\beta}^{+-} \hat{c}_\alpha^\dagger \hat{c}_\beta + h_{\alpha\beta}^{-+} \hat{c}_\alpha \hat{c}_\beta^\dagger \right] + (\text{same sign terms}) \\ &= \sum_{\alpha,\beta} \left[h_{\alpha\beta}^{+-} (\delta_{\alpha\beta} - \hat{c}_\beta \hat{c}_\alpha^\dagger) + h_{\alpha\beta}^{-+} (\delta_{\alpha\beta} - \hat{c}_\beta^\dagger \hat{c}_\alpha) \right] + (\text{same sign terms}) \\ &= - \sum_{\alpha,\beta} \left[h_{\beta\alpha}^{+-} \hat{c}_\alpha \hat{c}_\beta^\dagger + h_{\beta\alpha}^{-+} \hat{c}_\alpha^\dagger \hat{c}_\beta \right] + (\text{same sign terms}) \end{aligned}$$

In second row we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h_{\beta\alpha}^{+-} = -h_{\alpha\beta}^{-+} \quad h_{\beta\alpha}^{-+} = -h_{\alpha\beta}^{+-}$$

thus, also on-diagonal fields are linked. We are allowed to determine, say, just h^{+-} : its counterpart h^{-+} is completely determined.

2.2.2 Self-consistent variational solution

We follow and extend the variational scheme proposed by Fabrizio [6]. Let:

$$\Delta_{\alpha\beta}^{pq} \equiv \langle \hat{a}_\alpha^p \hat{a}_\beta^q \rangle$$

thus, explicitly

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \quad \Delta_{\alpha\beta}^{-+} = \langle \hat{c}_\alpha \hat{c}_\beta^\dagger \rangle \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \quad \Delta_{\alpha\beta}^{--} = \langle \hat{c}_\alpha \hat{c}_\beta \rangle$$

The first two are “normal densities”; the second two are “anomalous densities”. The HF ground-state energy is given by:

$$\begin{aligned} E_{\text{HF}} &= \langle \hat{H}_{\text{HF}} \rangle \\ &= \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \Delta_{\alpha\beta}^{pq} \end{aligned}$$

Here $\Delta_{\alpha\beta}^{pq}$ are functions of the fields $h_{\alpha\beta}^{pq}$. Derive energy:

$$\begin{aligned} \frac{\partial E_{\text{HF}}}{\partial h_{\mu\nu}^{rs}} &= \frac{\partial}{\partial h_{\mu\nu}^{rs}} \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \Delta_{\alpha\beta}^{pq} \\ &= \Delta_{\mu\nu}^{rs} + \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \frac{\partial \Delta_{\alpha\beta}^{pq}}{\partial h_{\mu\nu}^{rs}} \end{aligned}$$

From Hellman-Feynman theorem [8], we know that for a parametric hamiltonian it holds:

$$\begin{aligned}
\frac{\partial E_{\text{HF}}}{\partial h_{\mu\nu}^{rs}} &= \langle \Psi | \frac{\partial \hat{H}_{\text{HF}}}{\partial h_{\mu\nu}^{rs}} | \Psi \rangle \\
&= \langle \Psi | \frac{\partial}{\partial h_{\mu\nu}^{rs}} \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \hat{a}_{\alpha}^p \hat{a}_{\beta}^q | \Psi \rangle \\
&= \langle \hat{a}_{\mu}^r \hat{a}_{\nu}^s \rangle \\
&= \Delta_{\mu\nu}^{rs}
\end{aligned}$$

The last two equations imply:

$$\sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \frac{\partial \Delta_{\alpha\beta}^{pq}}{\partial h_{\mu\nu}^{rs}} = 0$$

This must hold for any possible choice of fields $h_{\alpha\beta}^{pq}$. Note that we did not impose the energy derivative to be zero: it is not the HF energy we are minimising.

Let us get back to the actual hamiltonian $\hat{H}$. To reduce it to a quadratic form is equivalent to assume that its ground-state has the properties described in previous sections. Then the following decomposition must hold:

$$\begin{aligned}
E &= \langle \hat{H} \rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \langle \hat{a}_{\alpha}^{+} \hat{a}_{\beta}^{-} \rangle + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\langle \hat{a}_{\alpha}^{+} \hat{a}_{\beta}^{+} \rangle \langle \hat{a}_{\gamma}^{-} \hat{a}_{\delta}^{-} \rangle - \langle \hat{a}_{\alpha}^{+} \hat{a}_{\gamma}^{-} \rangle \langle \hat{a}_{\beta}^{+} \hat{a}_{\delta}^{-} \rangle + \langle \hat{a}_{\alpha}^{+} \hat{a}_{\delta}^{-} \rangle \langle \hat{a}_{\beta}^{+} \hat{a}_{\gamma}^{-} \rangle \right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \Delta_{\alpha\beta}^{+-} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\Delta_{\alpha\beta}^{++} \Delta_{\gamma\delta}^{--} - \Delta_{\alpha\gamma}^{+-} \Delta_{\beta\delta}^{+-} + \Delta_{\alpha\delta}^{+-} \Delta_{\beta\gamma}^{+-} \right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \Delta_{\alpha\beta}^{+-} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \Delta_{\alpha\beta}^{+-} \Delta_{\gamma\delta}^{+-} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \Delta_{\alpha\beta}^{++} \Delta_{\gamma\delta}^{--}
\end{aligned}$$

Take the derivative:

$$\begin{aligned}
\frac{\partial E}{\partial h_{\mu\nu}^{rs}} &= \sum_{\alpha,\beta} A_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \left(\frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} \Delta_{\gamma\delta}^{+-} + \Delta_{\alpha\beta}^{+-} \frac{\partial \Delta_{\gamma\delta}^{+-}}{\partial h_{\mu\nu}^{rs}} \right) \\
&\quad + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\frac{\partial \Delta_{\alpha\beta}^{++}}{\partial h_{\mu\nu}^{rs}} \Delta_{\gamma\delta}^{--} + \Delta_{\alpha\beta}^{++} \frac{\partial \Delta_{\gamma\delta}^{--}}{\partial h_{\mu\nu}^{rs}} \right)
\end{aligned}$$

Using Eq. (2.3) we end up with:

$$\begin{aligned}
\frac{\partial E}{\partial h_{\mu\nu}^{rs}} &= \sum_{\alpha,\beta} \left[A_{\alpha\beta} + \sum_{\gamma,\delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \Delta_{\gamma\delta}^{+-} \right] \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} \\
&\quad + \sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\gamma\delta}^{--}}{2} \frac{\partial \Delta_{\alpha\beta}^{++}}{\partial h_{\mu\nu}^{rs}} + \sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\alpha\beta}^{++}}{2} \frac{\partial \Delta_{\gamma\delta}^{--}}{\partial h_{\mu\nu}^{rs}}
\end{aligned}$$

Let now:

$$C_{\alpha\beta} \equiv A_{\alpha\beta} + \sum_{\gamma,\delta} (B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}) \Delta_{\gamma\delta}^{+-} \quad D_{\alpha\beta} = \sum_{\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta_{\gamma\delta}^{--}}{2}$$

From Fermi anticommutation rules, we have:

$$\Delta_{\alpha\beta}^{+-} = \delta_{\alpha\beta} - \Delta_{\beta\alpha}^{-+} \implies \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} = - \frac{\partial \Delta_{\beta\alpha}^{-+}}{\partial h_{\mu\nu}^{rs}}$$

which implies:

$$\begin{aligned}\sum_{\alpha,\beta} C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} &= \frac{1}{2} \sum_{\alpha,\beta} \left(C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} + C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} \right) \\ &= \frac{1}{2} \sum_{\alpha,\beta} \left(C_{\alpha\beta} \frac{\partial \Delta_{\alpha\beta}^{+-}}{\partial h_{\mu\nu}^{rs}} - C_{\beta\alpha} \frac{\partial \Delta_{\alpha\beta}^{-+}}{\partial h_{\mu\nu}^{rs}} \right)\end{aligned}$$

[To be continued...]

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